Title: An analysis of 30 years of self-reported mountaineering accidents

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Intro/Background/Primary Objectives:

Outdoor recreation has increased in the United States in the last decade. Statistics on the number of Americans participating in climbing are kept by the American Alpine Club (AAC), a non-profit organization founded in 1902 to promote climbing and mountaineering. In 2014 the number of climbers across all disciplines (rock, ice, snow, indoor) was estimated at 7.7 million participants rising to 9.5 million in 2022 (1, 2). With increasing numbers of participants in potentially risky outdoor activities, accidents in remote and wilderness areas have become an increasing concern.

Efforts to compile accident data involve the AAC, National Park Service (NPS) incident data, and Mountain Rescue Association (MRA) search and rescue mission summaries. The AAC yearly report, *Accidents in North American Climbing (ANAC)*, represents the longest continuous dataset of mountain accidents in the United States, having been established in 1948. Data in ANAC is derived from self-submitted incident reports, search and rescue reports, NPS reports, and reports submitted by guiding services. The yearly report contains accidents from all climbing disciplines (rock, alpine, ice, mountaineering) and includes location data, date, cause of the accident, and contributing factors.

Mountaineering and climbing as disciplines have changed substantially since the inaugural ANAC report in 1948. Participation in climbing activities has increased, and the advent of the climbing gym has led to greater participation by relatively inexperienced individuals. There have been advances in safety gear such as the introduction of dynamic ropes and a normalization of helmet usage. There has been greater participation in wilderness medicine courses and self-rescue training among climbers. These changes have resulted in alterations in accident patterns.

We propose using a database of the narrative ANAC reports to investigate trends in mountaineering accidents over 30 years of data. Our primary objectives are:

- To describe trends in proximal accident cause over time (AAC categories include fall on rock, rappel error, avalanche, exposure, rock fall)
- To map the geographic distributions of mountaineering accidents

- To compare the number of accidents per discipline (rock, ice, alpine, mountaineering) over time against timelines of introductions of novel safety gear (for example camming devices for rock protection, dynamic ropes, helmets)
- To compare what proportion of accidents, result in a) self-rescue, b) professional rescue, c) hospitalization and whether there is a correlation with prior medical or rescue experience on the part of the participants.

Data Description and Acquisition:

Our dataset comes from a publicly available GitHub repository: https://github.com/ecaroom/climbing-accidents

The repository is openly accessible and licensed for public use. The data is stored in an excel file containing structured data that documents climbing accident incidents. Each row corresponds to a single recorded incident, and each column represents an attribute that describes the accident.

The dataset contains 2,770 rows, each representing a climbing accident. The first few columns in the dataset describe the accident, year, additional descriptive information, and tags that were applied to the accident. The additional attributes describing each accident includes:

- Date of incident
- Location
- Age Range
- Ability level
- Injury type
- Type of climbing
- Cause of the accident
- Weather/site conditions during the accident
- Equipment involved

There are additional tabs included in the dataset including the total incident count, roped count by style, and TR-sport-trad ex-ALP. Because the dataset includes location information, this allows for spatial analysis and different visualization techniques described below. The data includes dates and categorical variables that can allow us to compare and classify different features for our analysis.

The dataset is directly downloaded from GitHub. It does not require API access, web-scraping, or an account to access. Since the repository is publicly available, we are allowed to download and analyze the data for academic purposes.

Ethical Considerations:

This project directly addresses individuals partaking in mountaineering activities, including varied disciplines of climbing, skiing, and traversing. This dataset contains reported incidents in mountaineering, where individuals may be identified based on unique circumstances of the occurrence. However, these incidents were self-reported, so their consent was most likely required for publication. A harmful outcome of this project could be public perception on outdoor activities, where some might be discouraged in participating.

Alpine guides on gear, safety, and recommended practices might also be influenced by our project, where trends in incidents may reveal holes in the widely accepted principles. Popular organizations that publish include the American Alpine Club, International Climbing and Mountaineering Federation, and International Federation of Mountain Guides Association. These are globally renowned organizations that prioritize safe practices that reduce injuries in alpine settings, advising gear, route selection, and activity choice.

Companies that manufacture gear uphold to the highest standards, prioritizing safety and minimizing liability surrounding the brand and produced gear. These companies could be considered stakeholders, as improper user instruction or faulty mechanisms might influence the overlap between gear safety and operations. However, this project does not intend to influence perception on individual brands or companies, but if a product is mentioned in the report, it should be considered if it is listed as the reason for injury.

An additional stakeholder is medical professionals and search/rescue teams. This project could advise efficient responses, adequate training and funding, and the availability of resources. These are relevant to reactions but limited in preventing occurences.

Data Cleaning and Processing:

Our data has undergone some cleaning and organization prior to the project. We considered rebuilding the dataset from scratch to incorporate accident data after 2018, however the AAC has since disabled the website search function that made the initial data gathering possible.

Currently the data is housed in a CSV file with narrative reports of the accidents linked to descriptive tags. The author of this csv file has done the following data cleaning steps:

- Removal of duplicate entries

- Cleaning of aggregate reports (shared reports submitted by multiple climbers in the same incident)
- Tags for date, location, climbing type, age, experience, outcome severity

There is still substantial data cleaning and preparation to be done to complete our primary objectives. Most importantly, we will need to add our own tags to the narrative reports.

- To accurately map the locations of incidents, we will need to map each incident to GPS coordinates, and not a text-based descriptor.
- Although experience currently exists as a tag, there is no granularity in this descriptor, and we will add tags for formal experience (wilderness first aid, wilderness first responder, self-rescue training, etc).
- There is also no current tag for type of rescue. We propose tagging each incident as self-rescue or professional rescue, with an additional tag if hospitalization was required.

As we are dealing with narrative accounts, adding tags to the data will require string processing. One strategy would be rule-based keyword tagging. In this case we would define keyword dictionaries ("medical training" : ["WFA", "WFR", "WEMT", "wilderness first aid", "wilderness first responder", "wilderness EMT"]) and use these to add tags to the incidents. Another strategy would be term-frequency-inverse document frequency (TF-IDF) to extract keywords in a less biased way. Finally, we could potentially use large language model tagging to automatically assign tags to the incidents.

Our ultimate plan is to try 2-3 of these strategies on a training set and validate the results on a test set and choose the one that performs best at tagging.

<u>Exploratory Analysis:</u>

To better understand the patterns in mountaineering and climbing accidents, we will perform various exploratory analyses using geographical and categorical visualization techniques. These analyses will help us to identify trends, relationships, and any correlations that are relevant to our primary question/project statement,

Because the dataset includes location information, we will perform spatial analysis of accidents across North America. We will convert the descriptive, text-based locations into GPS/ Lat-Long coordinates to plot the accident locations on a map of North America using scatter plots. We will also create a heatmap of North America showing the accident density by region. To determine whether certain climbing disciplines, like rock, alpine, ice and mountaineering, differ or cluster in specific locations, the heat map can help

determine these correlations. This type of analysis will allow us to identify regions with high accident numbers and whether there is a correlation with location and accidents.

Another method we will perform is determining the temporal trends and finding changes over time. We will analyze the frequency of accidents and type over the years. We will create a histogram of the total recorded incidents by year. Line plots will also describe the trends in accidents over time. Seasonal histograms will help us identify the seasonal patterns in climbing accidents. These visualizations will allow us to determine whether the accident frequency is increasing or decreasing with time.

To investigate the correlations between equipment advancement and injury patterns, we will create a timeline plot overlying safety equipment introductions, such as helmets, carabiners, and other devices, with accident frequency each year. Compare the period before and after these instruments were introduced by displaying side-by-side bar charts. These visualizations may reveal correlations between safety advances and reduction in accidents.

To address our objective regarding rescue type and participant experience, we will create a bar chart comparing proportions of self-rescue, professional rescue, and hospitalization. Use will create bar charts that compare the rescues by experience level.

Our exploratory analysis will allow us to describe the analysis methodology that is further discussed in the Analysis Methodology section. These findings will help us describe our findings to the class as well as help us with further analytical problems in the project.

Analysis Methodology:

The preliminary data we collected contains individual incidents reported, involving location, activity, individuals, conditions, potential mistakes, method of rescue, and date of occurrence. We'd like to investigate rescues of individuals, including self-rescue, EMS, Search and Rescue, and hospitalization; this will allow us to understand how mountaineering incidents can be mitigated and the role of medical intervention. Additionally, we'd like to scrape the data analyzing how incidents occurred (i.e. equipment malfunctions, user errors, unfavorable weather conditions, and terrain failures). This can provide a more comprehensive idea of how and why injuries occur in mountaineering settings. An interesting analysis we'd like to include is a geographic distribution of incidents, identifying locations of events in the data relative to popular regions of climbing. Lastly, we'd like to look at a temporal analysis of incidents, where advancements in gear and mountaineering guidelines may reduce the risk of incidents.

Our methodology will include scraping and sorting incidents according to these parameters, where certain keywords and classifications of injury type/fatality, activity, location, and cause of incident can restructure the original dataset. This will allow for visualization of data and statistical analysis.

Preliminary Project Schedule:

Project Schedule		
Week of:	**Tasks:**	**Due**
3/16/2026	Work through testing 2-3 strategies to successfully add tags	Friday, March 20, 2026
3/23/2026	Begin exploring the dataset and begin exploratory analysis	Monday, March 23, 2026
3/23/2026	Write the "First Milestone" report showing our completed exploratory analyses upcoming methodolgy.	Friday, March 27, 2026
3/30/2026	Finalize visual analysis and complete other analyses	
3/13/2026	Finalize project submittal and begin video presentation	Tuesday, April 21, 2026

*The team will be in contact virtually or in person to discuss project tasks and timeline every week to ensure the project is on track to be completed on time.

References

1. American Alpine Club. (2019, July 9). *2019 State of Climbing report*. Retrieved from https://americanalpineclub.org/news/2019/7/8/2019-state-of-climbing-report
2. Rock Climbing Statistics: Market Data Report 2026. (2026). *Gitnux.org*. Retrieved February 2026, from https://gitnux.org/rock-climbing-statistics/